

Fuzzy Logic

Classic logic

In classical (binary) logic, a proposition can be either true or false. We may say that the truth value of a proposition P , $T(P)$, is a function which maps to either true (1) or false (0).

$$T(P) \rightarrow \{0, 1\}$$

Alternatively, we can think of propositions as a set X within a universe U .

$$T : X \subseteq U \rightarrow \{0, 1\}$$

Logical connectives

Connective	Expression	Simplified
Conjunction	$A \wedge B$	Logical AND
Disjunction	$A \vee B$	Logical OR
Negation	$\neg A$	Logical NOT
Implication	$A \rightarrow B$	If A is true, B cannot be false, i.e. $\neg A \vee B$
Equivalence	$A \leftrightarrow B$	$A \rightarrow B \wedge B \rightarrow A$

Sets and logic

Consider two sets A and B defined on universe X . We can define two propositions P and Q such that:

$$P : x \in A$$

$$Q : x \in B$$

In terms of truth value:

$$T(P) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

$$T(Q) = \begin{cases} 1 & \text{if } x \in B \\ 0 & \text{if } x \notin B \end{cases}$$

Note that $T(P) = \chi_A(x)$.

Using logical connectives

Conjunction	$T(P \wedge Q) = \min(T(P), T(Q))$
Disjunction	$T(P \vee Q) = \max(T(P), T(Q))$
Negation	$T(\neg P) = 1 - T(P)$
Implication	$T(P \rightarrow Q) = T(\neg P) \vee T(Q)$
Equivalence	$T(P \leftrightarrow Q) = T(P) = T(Q)$

Compound proposition

If proposition P and Q are defined on sets A and B respectively (A and B have universe X and Y respectively), then IF A , THEN B or $P \rightarrow Q$ can be denoted by the relation:

$$R = (A \times B) \cup (\overline{A} \times Y)$$

Similarly, IF A , THEN B , ELSE C , where C is also defined on universe Y , can be denoted by the relation:

$$R = (A \times B) \cup (\overline{A} \times C)$$

Tautology

Tautologies are compound propositions which are always true.

Some common tautologies

$$A \cup \overline{A} \leftrightarrow X$$

$$A \cup X; \overline{A} \cup X \leftrightarrow X$$

$$(A \wedge (A \rightarrow B)) \rightarrow B \quad (\text{modus ponens})$$

$$(\overline{B} \wedge (A \rightarrow B)) \rightarrow \overline{A} \quad (\text{modus tollens})$$

Proof of modus ponens

$$\begin{aligned}
& (A \wedge (A \rightarrow B)) \rightarrow B \\
& \neg(A \wedge (A \rightarrow B)) \vee B && (\text{implication}) \\
& \neg A \vee \neg(A \rightarrow B) \vee B && (\text{de Morgan's law}) \\
& \neg A \vee \neg(\neg A \vee B) \vee B && (\text{implication}) \\
& \neg A \vee (A \wedge \neg B) \vee B && (\text{de Morgan's law}) \\
& ((\neg A \vee A) \wedge (\neg A \vee \neg B)) \vee B && (\text{distributivity}) \\
& (\text{true} \wedge (\neg A \vee \neg B)) \vee B && (\text{domination law}) \\
& (\neg A \vee \neg B) \vee B && (\text{domination law}) \\
& \neg A \vee (\neg B \vee B) && (\text{associativity}) \\
& \neg A \vee \text{true} && (\text{domination law}) \\
& \text{true} && (\text{domination law})
\end{aligned}$$

Proof of modus tollens

$$\begin{aligned}
& (\neg B \wedge (A \rightarrow B)) \rightarrow \neg A \\
& \neg(\neg B \wedge (A \rightarrow B)) \vee \neg A && (\text{implication}) \\
& B \vee \neg(A \rightarrow B) \vee \neg A && (\text{de Morgan's law}) \\
& B \vee \neg(\neg A \vee B) \vee \neg A && (\text{implication}) \\
& B \vee (A \wedge \neg B) \vee \neg A && (\text{de Morgan's law}) \\
& ((B \vee A) \wedge (B \vee \neg B)) \vee \neg A && (\text{distributivity}) \\
& ((B \vee A) \wedge \text{true}) \vee \neg A && (\text{domination law}) \\
& (B \vee A) \vee \neg A && (\text{domination law}) \\
& B \vee (A \vee \neg A) && (\text{distributivity}) \\
& B \vee \text{true} && (\text{domination law}) \\
& \text{true} && (\text{domination law})
\end{aligned}$$

Contradiction

- # A contradiction is a proposition which is always false.
- # E.g. $A \cap \bar{A}$, $A \cap \phi$, $\bar{A} \cap \phi$, etc.

Fuzzy logic

- # Fuzzy logic deals with multivalued logic.
- # A fuzzy logic proposition, P , is a statement without clearly defined boundaries.
- # The truth value, $T(P)$, maps to any real value from $[0, 1]$.

$$T : X \subseteq U \rightarrow [0, 1]$$

- # Fuzzy propositions are associated to fuzzy sets.

$$\begin{aligned} P : x \in A \\ \Rightarrow T(P) = \mu_A(x) \mid 0 \leq \mu_A(x) \leq 1 \end{aligned}$$

Approximate reasoning

- # Approximate reasoning is used to infer information from imprecise propositions.
- # It is an extension of predicate calculus. Predicate calculus reasons using precise propositions, where as approximate reasoning is suitable even for vague and half truths.
- # If we are given a fuzzy linguistic proposition of the form IF A THEN B , then we can find the relation between A and B , with their universe of discourse X and Y respectively, similar to how we did in the case of classical logic:

$$R = (A \times B) \cup (\bar{A} \times Y)$$

After that, if we are given a new set of antecedents A' , then we can find the new set of consequences B' by using composition:

$$B' = A' \circ R$$